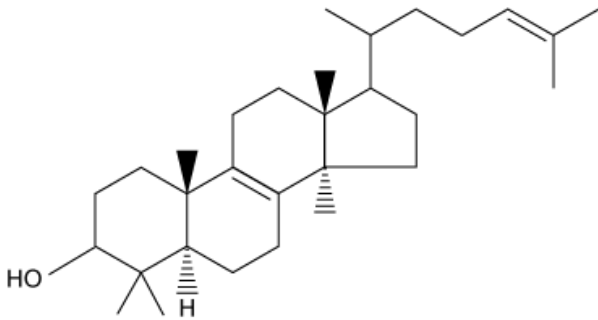


Terpenes are synthesized from two or more five-carbon groups known as isoprenes. The triterpenoid lanosterol, a steroid precursor, is shown below.



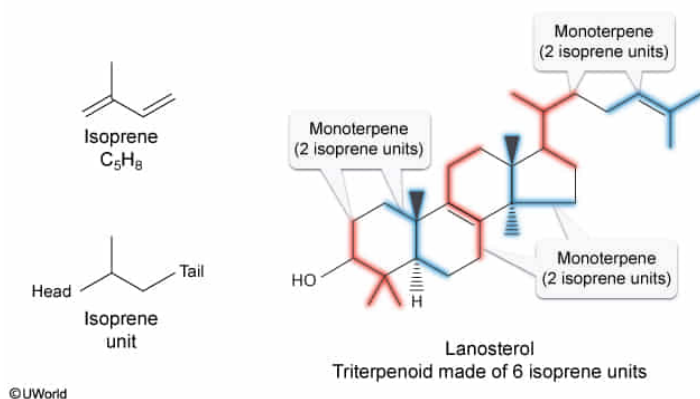
How many isoprene units are present in lanosterol?

- ☐ A. 3 [27%]
- ☐ B. 4 [22%]
- ☒ C. 6 [46%]
- ☐ D. 8 [3%]

Correct

47% answered correctly

Explanation:



Terpenes are lipid precursors to steroids and lipid signaling molecules. These compounds are made up of branched five-carbon groups called **isoprene units**, in which the branched end is referred to as the head and the unbranched end is the tail. Isoprene units can be joined to

form terpenes in one of three ways: **head-to-tail, tail-to-tail, or head-to-head**.

Terpenes are categorized according to **how many isoprenes** are included. For example, the most basic terpene is made up of two isoprene units (10 carbon atoms total) and is known as a monoterpene. A diterpene contains double the number of isoprene units and carbon atoms as a monoterpene, giving a diterpene four isoprene units and 20 carbon atoms. Molecules made of isoprene units that also contain nonhydrocarbon functional groups are called terpenoids.

Lanosterol is a terpenoid synthesized during **cholesterol synthesis in vertebrates**. The question states that lanosterol is a **triterpenoid**, which can be thought of as three monoterpenes put together. Therefore, it is composed of **six isoprene units** ($3 \text{ terpenes} \times 2 \text{ isoprenes/terpene} = 6 \text{ isoprenes}$) and has **30 carbon atoms** in total ($6 \text{ isoprenes} \times 5 \text{ carbons/isoprene} = 30 \text{ carbons}$).

(Choice A) A triterpene can be thought of as three monoterpenes put together, which would contain six isoprenes rather than three. A terpene with three isoprenes is known as a sesquiterpene (ie, 1.5 monoterpenes).

(Choices B and D) Terpenes with four and eight isoprenes are known as diterpenes and tetraterpenes, respectively.

Educational objective:

Terpenes are a type of lipid made up of branched five-carbon units known as isoprenes. Terpenes are classified according to the number of isoprene units present in the molecule.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401525

System:
Functional Groups and Their Reactions